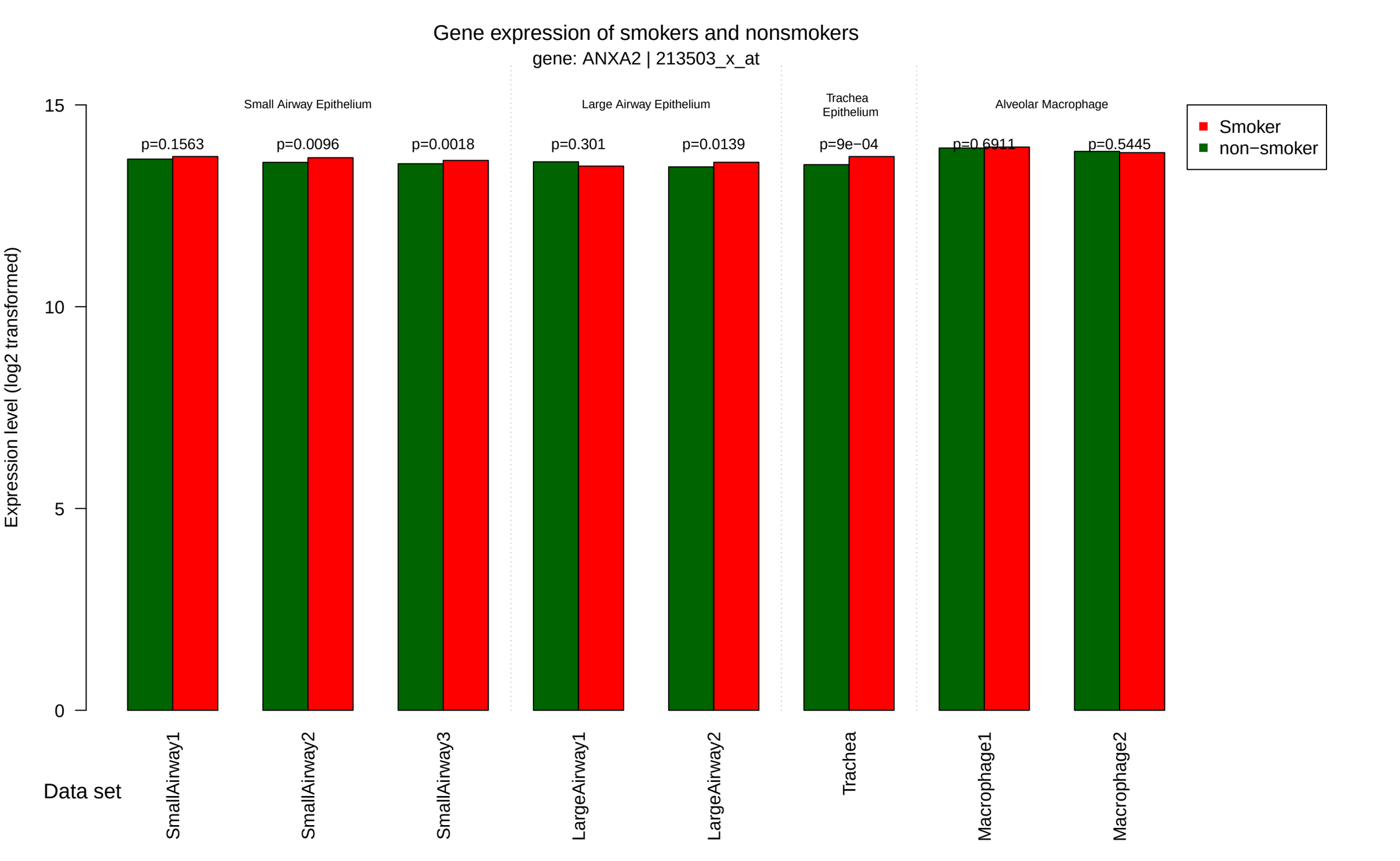
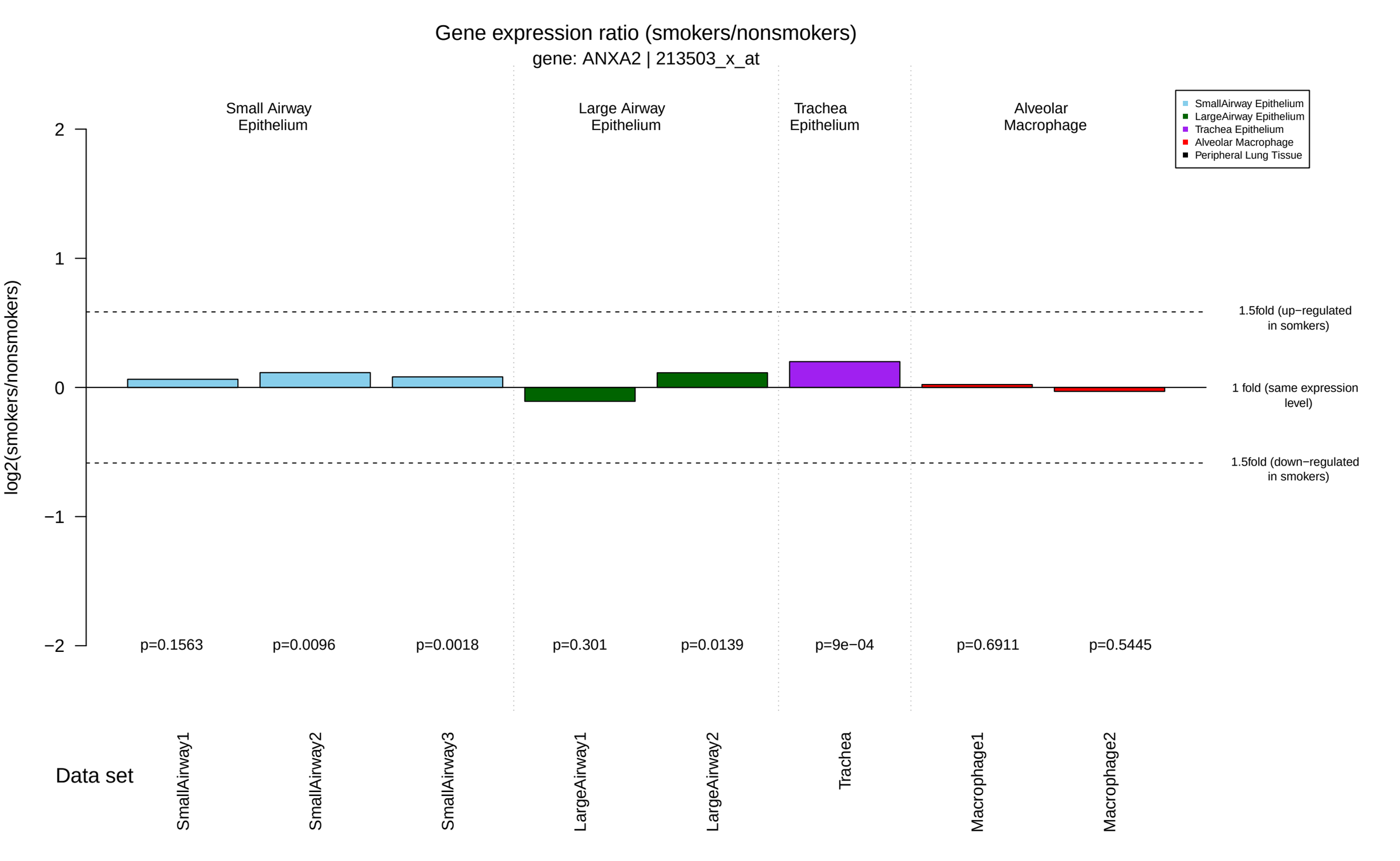
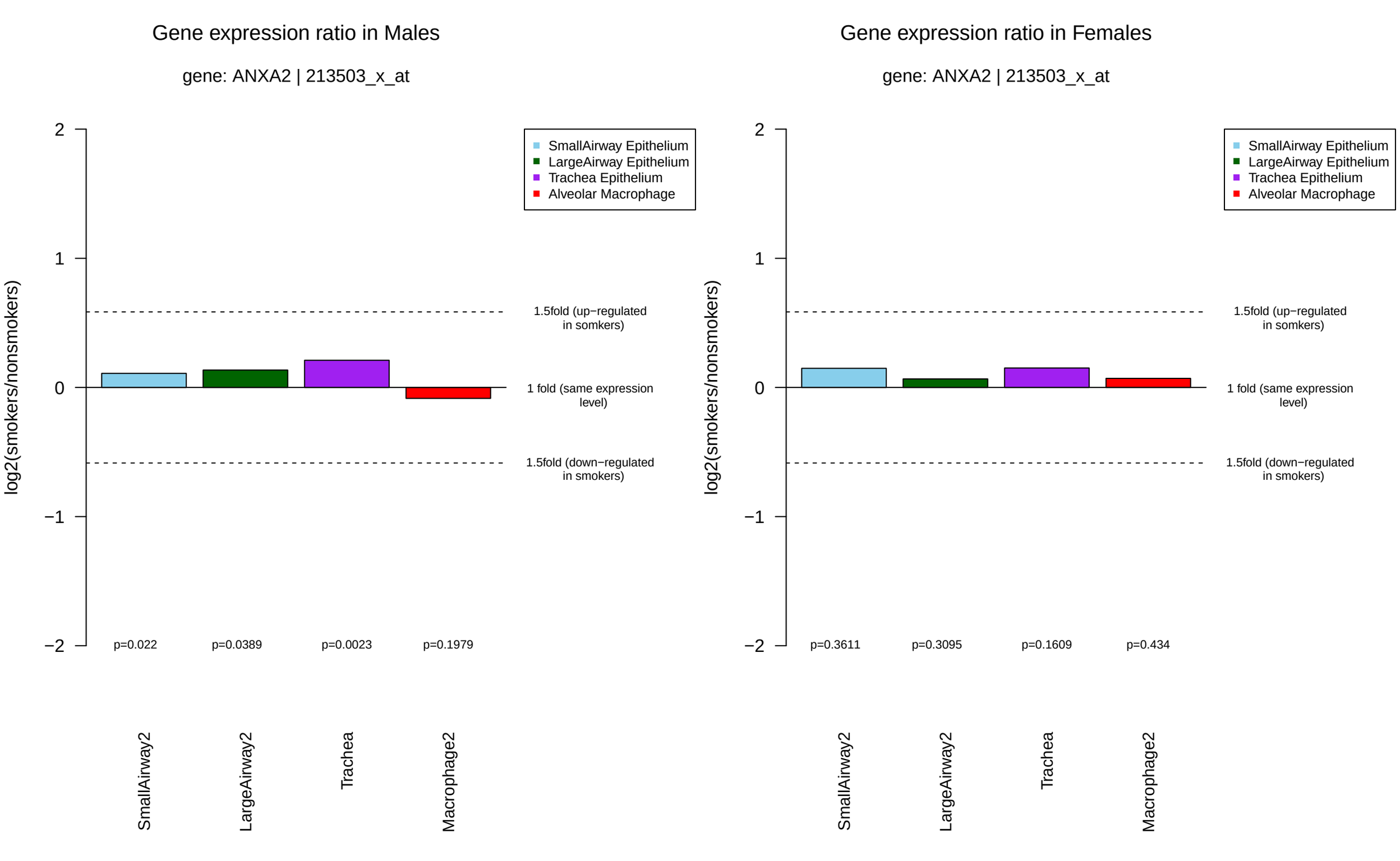
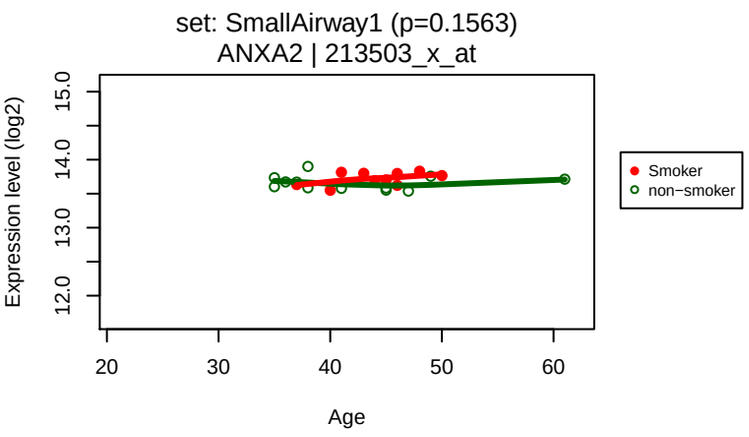
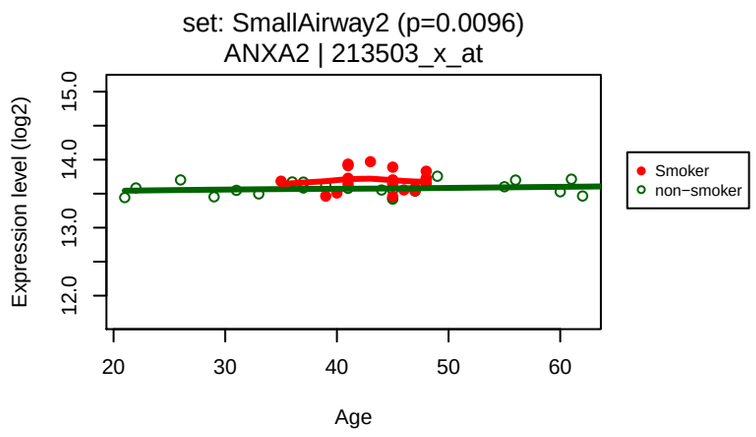
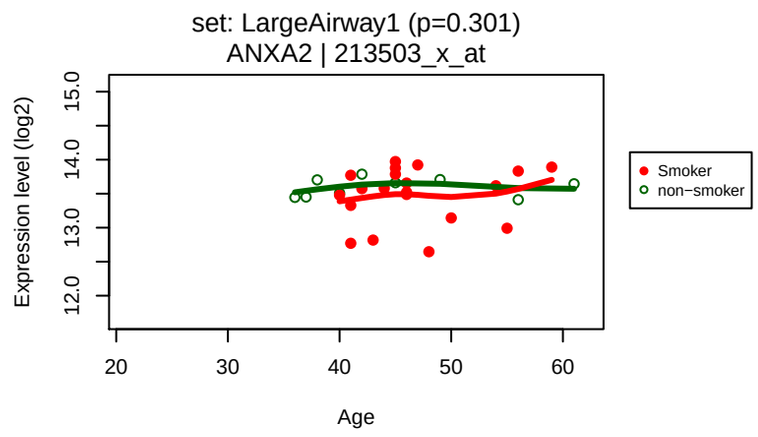

Gene expression of smokers and nonsmokers
gene: ANXA2 | 213503_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1563
p=0.0096
p=0.0018
p=0.301
p=0.0139
p=9e−04
p=0.6911
p=0.5445
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

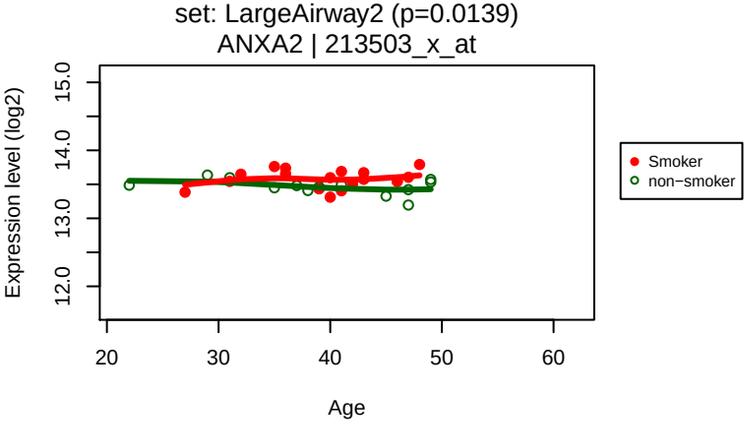

Gene expression ratio (smokers/nonsmokers)
gene: ANXA2 | 213503_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1563
p=0.0096
p=0.0018
p=0.301
p=0.0139
p=9e−04
p=0.6911
p=0.5445
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

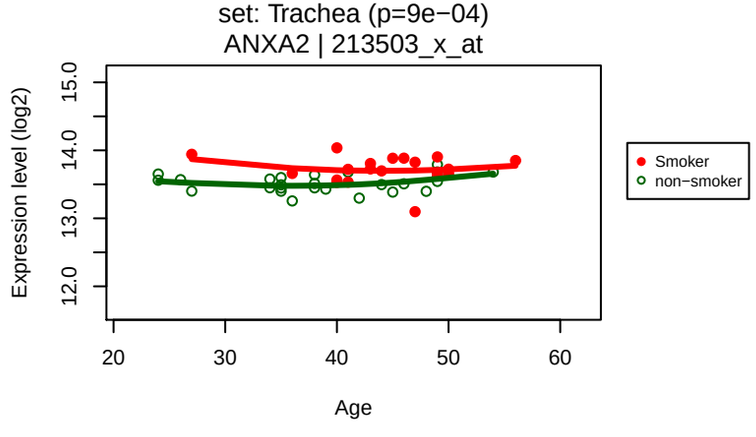

Gene expression ratio in Males
gene: ANXA2 | 213503_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.022
p=0.0389
p=0.0023
p=0.1979
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANXA2 | 213503_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3611
p=0.3095
p=0.1609
p=0.434
SmallAirway2
LargeAirway2
Trachea
Macrophage2

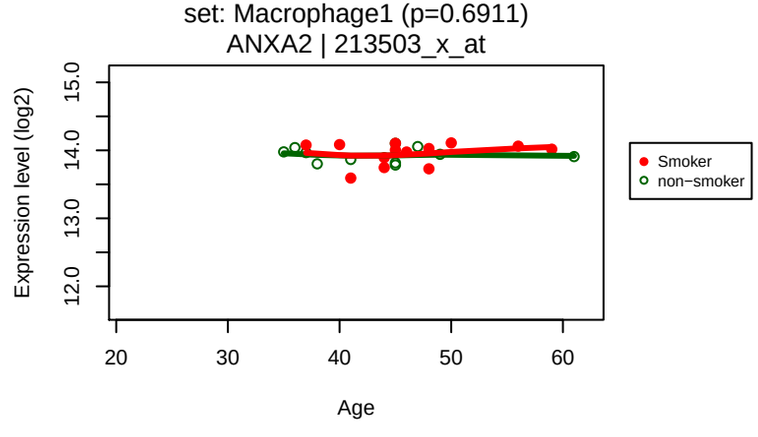

set: SmallAirway1 (p=0.1563)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

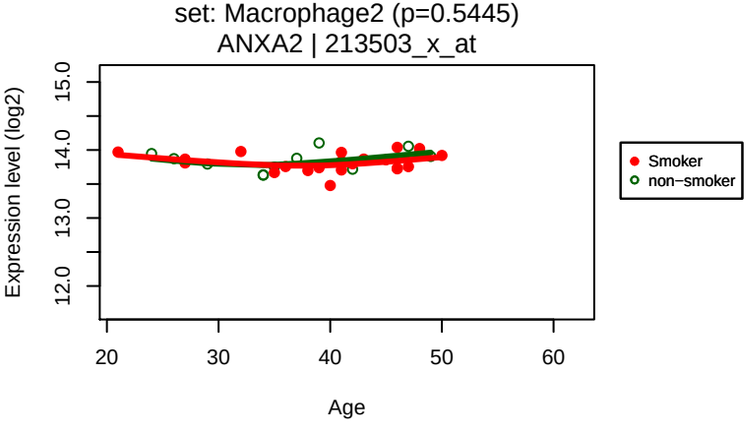

set: SmallAirway2 (p=0.0096)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

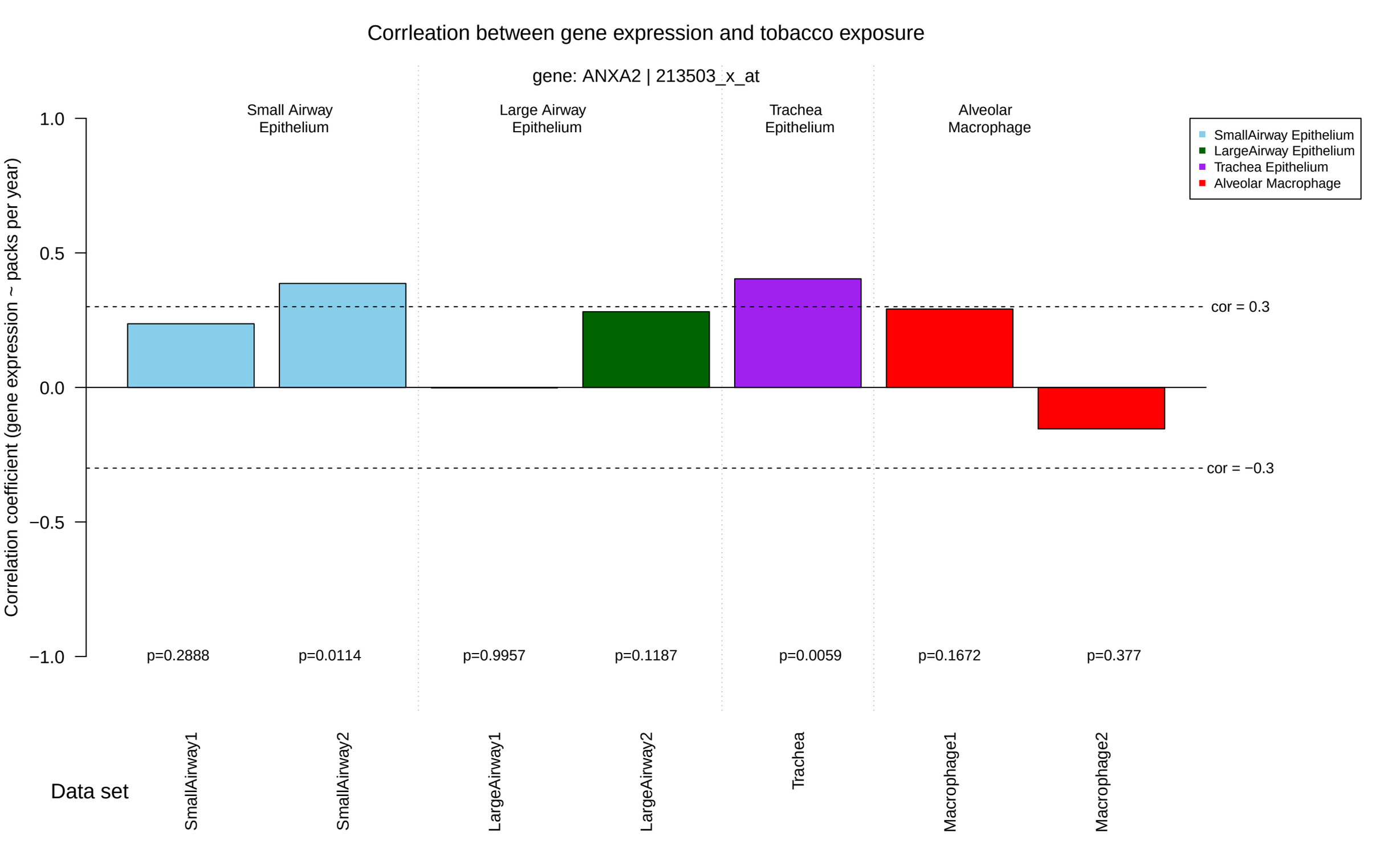

set: LargeAirway1 (p=0.301)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0139)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=9e−04)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6911)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5445)
ANXA2 | 213503_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANXA2 | 213503_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2888
p=0.0114
p=0.9957
p=0.1187
p=0.0059
p=0.1672
p=0.377
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2